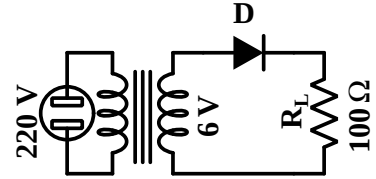


Tutorial-5 (Diode circuits and Applications)

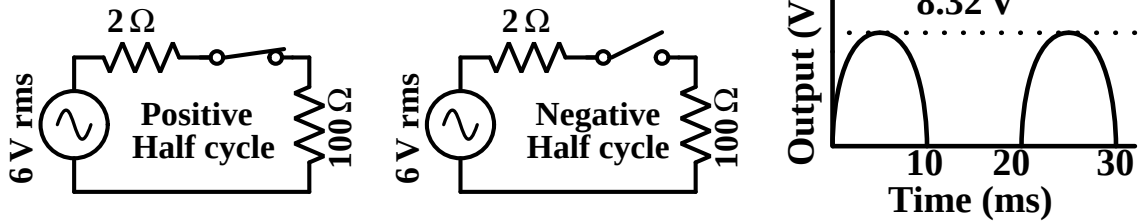
Q-1 Consider a diode being used as a rectifier.

A transformer reduces the 220 V 50 Hz mains voltage to 6 V. A diode connects the top of the secondary coil to the load resistor of 100 Ω . We can represent the transformer along with the resistance of the windings by an AC source of 6 V rms with a series resistor of 2 Ω .



- a) Using the ideal diode model, draw the equivalent circuit of the rectifier circuit above in the positive and negative half cycles. Draw the waveform of the voltage across the load resistor.

Soln. 1-a) The figures below show the equivalent circuits in positive and negative half cycles, as well as the output wave form.



- b) What is the peak voltage across the resistor? Compute the average voltage across the resistor over one cycle of AC.

Soln. 1-b) The peak voltage at the transformer secondary is $6\sqrt{2}$ V.

The output voltage is through a potential divider formed by the 2 Ω and 100 Ω resistors. Therefore peak output voltage is

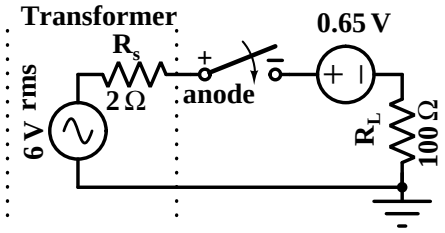
$$V_p = 6\sqrt{2} \frac{100}{2 + 100} = \frac{100\sqrt{2}}{17} \approx 8.32 \text{ V}$$

The output contains only the positive half of the AC cycle. The time period of the 50 Hz AC voltage is $= 1/50 = .02$ s. The output is $V_p \sin(100\pi t)$ for the first 10 ms of the cycle and 0 during the next 10 ms. Therefore the average value is:

$$\begin{aligned} V_{ave} &= \frac{1}{.02} \int_0^{.01} V_p \sin(100\pi t) dt = -\frac{V_p}{.02 \times 100\pi} \cos(100\pi t) \Big|_0^{.01} \\ &= -\frac{V_p}{2\pi} (\cos \pi - \cos 0) = -\frac{V_p}{2\pi} (-1 - 1) = \frac{V_p}{\pi} = \frac{8.319}{3.14159} = 2.648 \text{ V} \end{aligned}$$

- c) We now use the constant forward drop model for the diode.

Assume that the diode is equivalent to a switch with a series connected source of 0.65 V which subtracts from the input in the positive half cycle as shown in the figure. The switch is “on” only when the voltage across it (*the switch*) is positive.



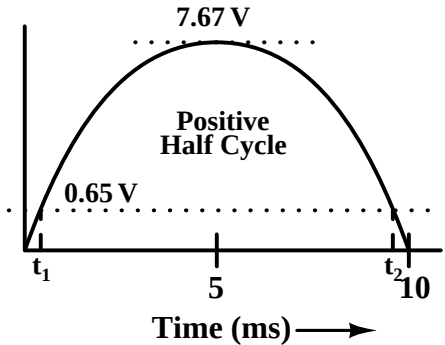
Does the diode conduct during the entire positive half-cycle of AC?

What is the peak voltage across the load resistor?

What is the average voltage across the resistor?

Soln. 1-c) Due to the 0.65 V source, the switch will close only when the input voltage rises above 0.65 V (and not 0 V) and will open as soon as the input voltage drops below 0.65 V.

Conduction takes place for less than half the cycle. There is no conduction (in the positive half cycle) till the input voltage rises to 0.65 V (at t_1) and no conduction (still in the positive half cycle) after it drops below 0.65 V (at t_2). The peak voltage reached is now $8.3189 - 0.65 = 7.6689 \approx 7.67$ V



The input rises to 0.65 V when $6\sqrt{2} \sin(100\pi t) = 0.65$. This happens when the phase angle $\theta_1 = 100\pi t = \sin^{-1}(0.0766) = 0.07668$ radians = 4.39° . The time when this phase angle is reached is: $t_1 = 0.07668/(100\pi) = 0.2441$ ms

Similarly, conduction stops when the phase angle is $\pi - 0.07668 = 3.065$ radians or $180 - 4.39 = 175.61^\circ$. The time when this phase angle is reached is $10 - 0.2441 = 9.7562$ ms.

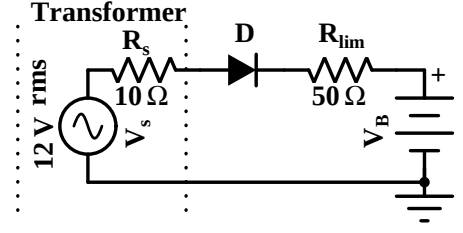
The average voltage across the resistor is given by

$$\begin{aligned}
 V_{ave} &= \frac{1}{T} \int_{t_1}^{t_2} (V_p \sin(100\pi t) - 0.65) dt \\
 &= \frac{1}{100\pi T} \int_{\theta_1}^{\theta_2} (V_p \sin(\theta) - 0.65) d\theta = \frac{1}{2\pi} \left(-V_p \cos \theta \Big|_{\theta_1}^{\theta_2} - 0.65(\theta_2 - \theta_1) \right) \\
 &= -\frac{V_p}{2\pi} (\cos \theta_2 - \cos \theta_1) - \frac{0.65}{2\pi} (\theta_2 - \theta_1) \\
 &= -\frac{8.319}{2\pi} (-0.9971 - 0.9971) - \frac{0.65}{2\pi} (3.065 - 0.07668) \\
 &= 2.6402 - 0.3091 = 2.331 \text{ V}
 \end{aligned}$$

If we ignore the shortening of the forward conduction time and just subtract 0.65 V from the peak voltage, we would have got $(8.319 - 0.65)/\pi = 2.441$ V as average, which is not a bad approximation.

Q-2 In the battery-charger circuit shown below, the transformer secondary is modeled as an AC source which provides 12 V rms with an effective series resistance of $R_s = 10 \Omega$.

The battery voltage V_B varies from 12 V to 14 V from the discharged to fully charged state. Consider D as an ideal diode. R_{lim} is a 50 Ω current limiting resistance placed by the designer.



Sketch and label the diode current waveform when $V_B = 12$ V.

What are its peak and average values?

What are the peak and average diode currents when $V_B = 14$ V?

Soln. 2-c) The peak voltage reached at the transformer secondary is $12\sqrt{2} = 16.97$ V.

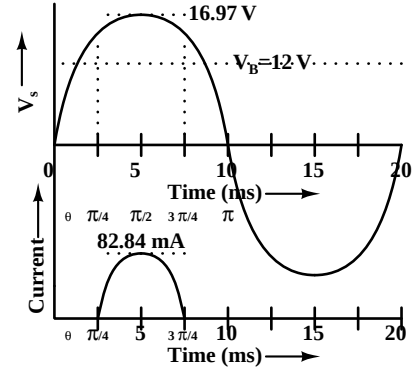
The source voltage can be represented by $V_s = 12\sqrt{2} \sin(2\pi \times 50 \times t) = 16.97 \sin(100\pi t)$, assuming the mains frequency to be 50 Hz.

The ideal diode will conduct when the transformer secondary voltage $V_s > 12$ V.

Let us define $\theta \equiv 100\pi t$. Then the diode will conduct when

$$12\sqrt{2} \sin \theta > 12 \quad \text{or} \quad \sin \theta > \frac{1}{\sqrt{2}}$$

Thus the conduction angle lies between 45° and 135° . Total resistance in the path is $10 + 50 = 60 \Omega$. When the diode conducts, the charging current is given by



$$(12\sqrt{2} \sin \theta - 12)/(10 + 50) = \frac{12}{60} (\sqrt{2} \sin \theta - 1) = 0.2 (\sqrt{2} \sin \theta - 1)$$

The peak charging current will be $0.2 \times (\sqrt{2} - 1) = 82.84$ mA. The average current will be given by

$$\begin{aligned} I_{ave} &= \frac{1}{2\pi} \int_{\pi/4}^{3\pi/4} 0.2 (\sqrt{2} \sin \theta - 1) d\theta = \frac{0.2\sqrt{2}}{2\pi} (-\cos \theta) \Big|_{\pi/4}^{3\pi/4} - \frac{0.2}{2\pi} (3\pi/4 - \pi/4) \\ &= \frac{0.1\sqrt{2}}{\pi} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) - \frac{0.2}{2\pi} \frac{\pi}{2} = \frac{0.2}{\pi} - \frac{0.1}{2} = .06366 - 0.05 = 0.01366 \text{ A} = 13.66 \text{ mA} \end{aligned}$$

If the battery voltage is 14 V, Charging current is given by

$$(12\sqrt{2} \sin \theta - 14)/(10 + 50) = 0.2\sqrt{2} \sin \theta - 0.23333, \quad I_{peak} = (0.2\sqrt{2} - 0.23333) = 49.51 \text{ mA}$$

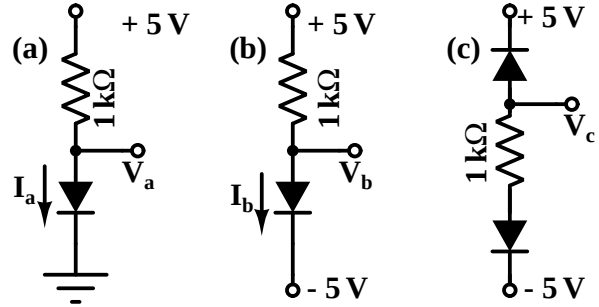
Onset of conduction is given by $12\sqrt{2} \sin \theta_1 = 14$, so $\sin \theta_1 = 14/(12\sqrt{2}) = 0.82496$. Therefore $\theta_1 = 0.9701$ radians = 55.58° .

Conduction will terminate at $\theta_2 = \pi - \theta_1 = 2.1715$ radians = 124.42° .

$$\begin{aligned} I_{ave} &= \frac{1}{2\pi} \int_{\theta_1}^{\theta_2} (0.2\sqrt{2} \sin \theta - 0.23333) d\theta = \frac{0.2\sqrt{2}}{2\pi} (-\cos \theta) \Big|_{\theta_1}^{\theta_2} - \frac{0.23333}{2\pi} (\theta_2 - \theta_1) \\ &= \frac{0.1\sqrt{2}}{\pi} (\cos \theta_1 - \cos \theta_2) - \frac{0.23333}{2\pi} (2.1715 - 0.9701) \\ &= \frac{0.1\sqrt{2}}{\pi} (0.5652 + 0.5652) - 0.03714 (2.1715 - 0.9701) = 6.27 \text{ mA} \end{aligned}$$

Q-3 In the three circuits (a), (b) and (c) shown below,

- a) find V_a , I_a , V_b , I_b , V_c , I_c using the ideal diode model.
- b) find V_a , I_a , V_b , I_b , V_c , I_c using the constant V_d model with $V_d = 0.7$ V.



Soln. 3) With the ideal diode model:

In circuit (a): The diode is forward biased, so $V_a = 0$ V.

$$I_a = 5 \text{ V} / 1 \text{ k}\Omega = 5 \text{ mA}$$

In circuit (b): The diode is forward biased, so $V_b = -5$ V.

$$I_b = (5 - (-5)) \text{ V} / 1 \text{ k}\Omega = 10 \text{ mA}$$

In circuit (c): The upper diode is blocked while the lower diode is forward biased. So $V_c = -5$ V, $I_c = 0$.

With constant drop model: $V_d = 0.7$ V

In circuit (a): The diode is forward biased, so $V_a = 0.7$ V.

$$I_a = (5 - 0.7) \text{ V} / 1 \text{ k}\Omega = 4.3 \text{ mA}$$

In circuit (b): The diode is forward biased, so $V_b = -4.3$ V.

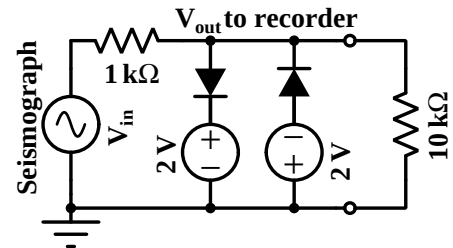
$$I_b = (5 - (-4.3)) \text{ V} / 1 \text{ k}\Omega = 9.3 \text{ mA}$$

In circuit (c): The upper diode is blocked so no current flows. $V_c = -5$ V, $I_c = 0$.

Q-4 A seismograph produces an output voltage representing ground movements due to earthquakes etc. Its output is a triangular wave which goes from $-V_p$ to V_p and back, where V_p is the peak voltage. Normally its output is confined to a peak voltage of < 1 V. Its output goes to a recording instrument with an input impedance of $10 \text{ k}\Omega$ which can accept voltages up to 2 V. The recording pen will be thrown out of range and get damaged if the input voltage exceeds 3 V.

The seismograph and the recorder are stationed in a laboratory close to an armament testing facility. In rare cases during testing of explosives at the nearby facility, the output of the seismograph may shoot up to 5 V peak. We propose to use the protection circuit shown on the right.

Assume an ideal diode model.

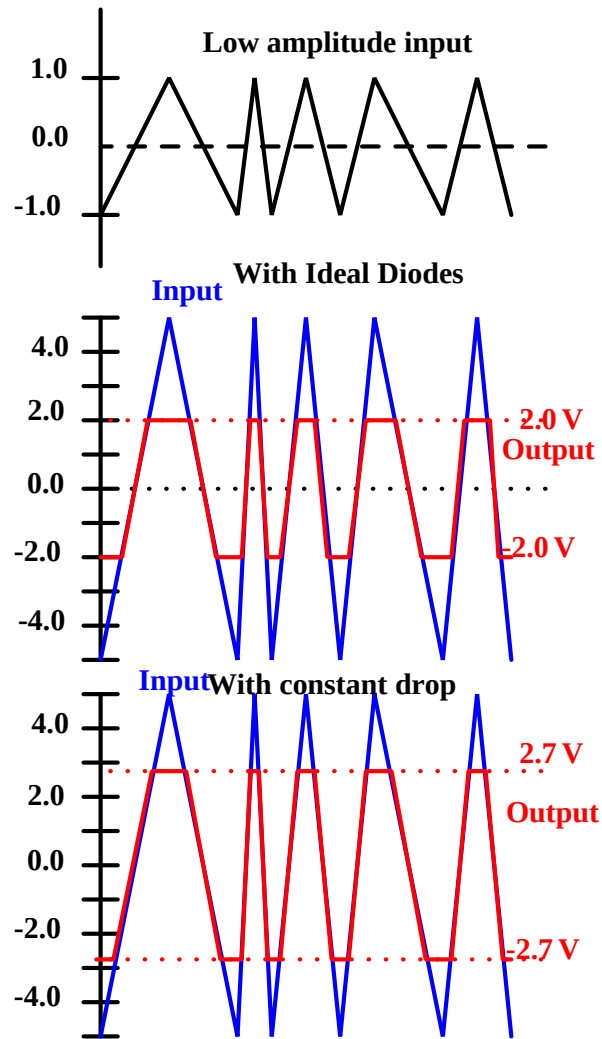


a) Plot the output waveshape in the normal case when $V_p = 1$ V.

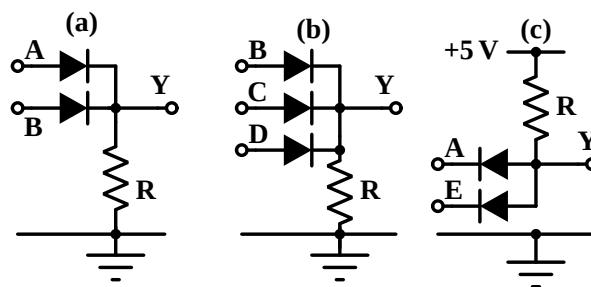
b) Plot the output waveshape when $V_p = 5$ V.

c) How will the output look if the diode is represented by a constant V_d model, with $V_d = 0.7$ V?

Soln. 4) The diagram below shows the three cases.



Q-5 Diodes can be used to implement logic gates. In the three circuits shown below, use the ideal diode model.



- a) What is the voltage at Y in circuits (a), (b) and (c) if $V_A = V_E = 5 \text{ V}$ and $V_B = V_C = V_D = 0 \text{ V}$?

Soln. 5-a)

Circuit (a): $V_A = 5 \text{ V}$, so upper diode is ON. $V_Y = 5 \text{ V}$.

Circuit (b): $V_B = V_C = V_D = 0$. None of the diodes conducts. $V_Y = 0 \text{ V}$.

Circuit (c): $V_A = V_E = 5 \text{ V}$. Neither diode conducts. $V_Y = 5 \text{ V}$.

- b) If 5 V represents logic '1', while 0 V represents logic '0', what is the logic function performed by the three circuits?

Soln. 5-b)

Circuit (a): $V_Y = 5 \text{ V}$ (= '1') if either or both of V_A and V_B are at 5 V (= '1'). So this is OR logic. $Y = A + B$.

Circuit (b): $V_Y = 5 \text{ V}$ (= '1') if one or more of V_B , V_C and V_D is at 5 V (= '1'). So this is OR logic. $Y = B + C + D$.

Circuit (c): Both V_A and V_E should be at 5 V for V_Y to be at 5 V. If either is at 0 V, $V_Y = 0 \text{ V}$. This is AND logic. $Y = A \cdot E$.

- c) If 5 V represents logic '0', while 0 V represents logic '1', what is the logic function performed by the three circuits?

Circuit (a): $V_Y = 5 \text{ V}$ (=logic '0') if either or both of V_A and V_B are at 5 V (= logic 0). Y is at 0 V (logic 1) only if both of A and B are at 0 V (logic 1). So this is AND logic. $Y = A \cdot B$.

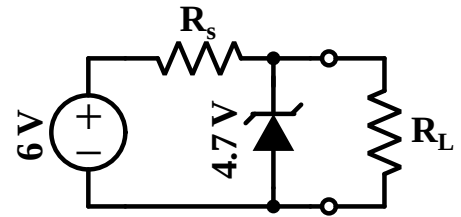
Circuit (b): $V_Y = 5 \text{ V}$ (=logic '0') if one or more of V_B , V_C and V_D are at 5 V (= logic 0). Y is at 0 V (logic 1) only if all of B, C and D are at 0 V (logic 1). So this is also AND logic. $Y = B \cdot C \cdot D$.

Circuit (c): $V_Y = 0 \text{ V}$ (=logic '1') if either or both of V_A and V_E are at 0 V. So this is OR logic. $Y = A + E$.

- Q-6** In the reverse bias breakdown mode, the current through the diode can change over a wide range, while the voltage changes very little. This phenomenon is used to generate stable voltages.

We have a DC voltage source which supplies 6 V. We want to supply power to a circuit using this source, where the supply voltage to the circuit should be kept as close to 4.7 V as possible.

The circuit to be powered can be represented as a load resistance R_L which can vary between 1 k Ω to 10 k Ω . We use a reverse biased diode whose breakdown voltage is 4.7 V, in the circuit configuration shown on the right. Recall that in breakdown region, the current through the diode can vary over several orders of magnitude with very little change in the voltage across it.



- a) What should be the value of the resistance R_s , such that the voltage across the load resistance is 4.7 V with the breakdown current through the diode = 1 mA when the load resistance is 1 k Ω .

Soln. 6-a) $R_L = 1 \text{ k}\Omega$, so current through it = 4.7 mA

Current through the Avalanche diode is 1 mA. So total current through R_s = 5.7 mA

Therefore $R_s = (6 - 4.7)/5.7 \times 10^{-3} = 228 \Omega$.

- b) What is the current through the diode when the load resistance is 10 k Ω .

Soln. 6-b) Current through $R_L = 4.7/10^4 = 0.47 \text{ mA}$. Current through $R_s = (6 - 4.7)/228 = 5.7 \text{ mA}$.

Therefore current through the Avalanche diode (KCL) is $5.7 - 0.47 = 5.23 \text{ mA}$.

- c) What is the worst case power dissipation in R_s and the diode for values of R_L between $1\text{ k}\Omega$ and $10\text{ k}\Omega$.

Soln. 6-c) Because R_s has a fixed voltage across it, power dissipation through it is always $(6 - 4.7)^2/228 = 7.41\text{ mW}$.

Voltage across the diode is 4.7 V , current through it is maximum when the current through the load is minimum. ($I_d = I_s - I_L$)

Current through the load is minimum when $R_L = 10\text{ k}\Omega$.

In that case, current through the diode is 5.23 mA . So power dissipation in the diode is $4.7 \times 5.23 \times 10^{-3} = 24.58\text{ mW}$.

(When a diode is used in this way, it is called a Zener diode. In fact, that is a misnomer because the breakdown current in most diodes used for such applications is due to avalanche multiplication and not due to Zener tunneling effect. However, the name has stuck and is widely used. Indeed, the symbol used for the diode when employed for such an application resembles the letter Z as seen in the diagram above.)